

Project Design Phase Solution Architecture

Date	22 July 2026
Team ID	TEAM-HT-2026
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

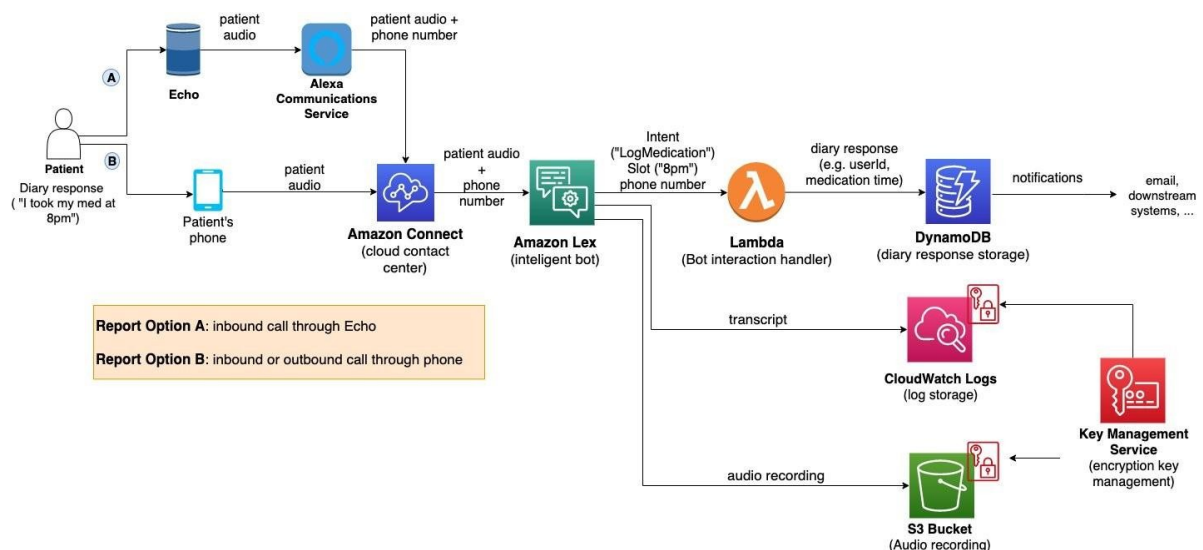


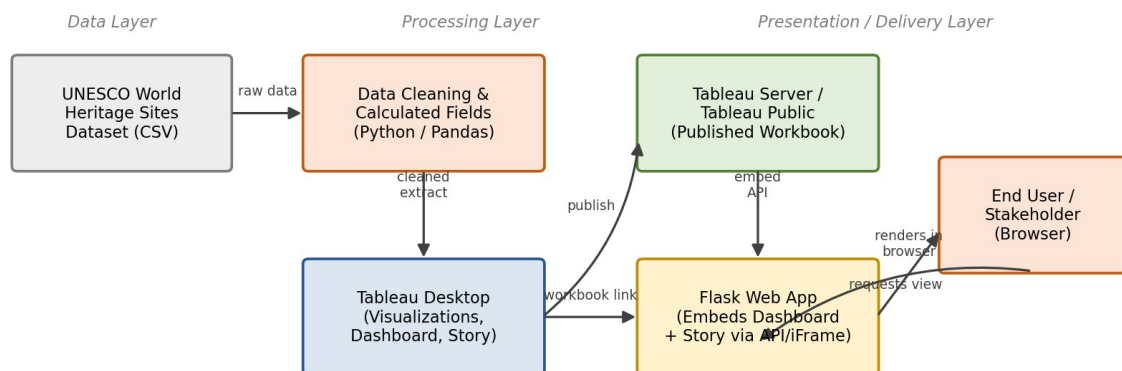
Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Our Solution Architecture – Heritage Treasures Project

The architecture below is organized into three layers: a Data Layer holding the raw UNESCO dataset, a Processing Layer that cleans the data and builds the Tableau visualizations, and a Presentation/Delivery Layer that publishes the workbook and embeds it in a Flask web app for end users.

Heritage Treasures - Solution Architecture



Architecture Components

Layer	Component	Role
Data Layer	UNESCO World Heritage Sites dataset (CSV)	Source of truth: Country, Name_en, Region, Danger, Date_inscribed and other attributes
Processing Layer	Python / Pandas cleaning script	Removes nulls/duplicates, standardizes names, prepares the file for Tableau
Processing Layer	Tableau Desktop	Builds calculated fields, the three core visualizations, dashboard and story
Presentation Layer	Tableau Server / Tableau Public	Hosts the published workbook and serves it via an embeddable link/API
Presentation Layer	Flask Web App	Embeds the dashboard/story into a branded web page and routes user requests
Presentation Layer	End User Browser	Displays the interactive dashboard/story to analysts,

		policy makers and researchers
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